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PI Name:	Hyde, Truell Ph.D.		
Project Title:	PK-4: Self-Ordering of Interacting Complex Plasma Particles in Microgravity		
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Project Type:	Flight	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
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COI Name (Institution):	Matthews, Lorin Ph.D. (Baylor University) Hartmann, Peter Ph.D. (Baylor University) Petrov, Oleg Ph.D. (Joint Institute for High Temperatures, Russian Academy of Sciences) Rosenberg, Marlene Ph.D. (University Of California, San Diego)		
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Task Description:	Complex plasma consists of ions, electrons, neutral gas, and solid microparticles referred to as dust. Dust grains immersed in plasma become negatively charged and interact with one another as well as the plasma components through electromagnetic interactions. In particular, dust grains experience a plasma drag force, altering the trajectories of ions in the plasma, and participate in collective plasma interactions, such as waves and instabilities. An interesting aspect of complex plasma is its ability to self-organize into dust liquids, 2D and 3D dust crystal lattices, and 1D dust chains, which are aligned with an electric field in the plasma. This feature provides researchers the ability to examine classical collective phenomena on size and time scales allowing for direct observation (at the kinetic level) of fundamental microscopic and macroscopic processes, as the dust can easily be imaged with high-speed cameras. In many environments, positive ions in the plasma have a directed flow with respect to the negatively charged dust grains. The resulting interaction between the dust and flowing plasma creates an ion wakefield downstream from the dust particles, with the positive space region modifying the interaction between the charged grains and contributing to the observed dynamics and particle structure. Since the ion wakefield is weak, the effects are usually masked by gravity for terrestrial experiments. This international research collaboration will use the PK-4 on the International Space Station (ISS) to conduct experiments in microgravity which investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains. Experiments are proposed to 1) determine the experimental conditions necessary to form stable, field-aligned chains, 2) study the interaction between adjacent chains and determine its impact on their structural and dynamic properties, and 3) examine the wake-mediated interaction forces between grains. Data obtained from orbit will be compared with data from two different ground-based experiments: a stratified and/or homogeneous DC glow discharge and a GEC RF reference cell with a glass box placed on the lower electrode, used to confine the particles in a vertical chain. Numerical modeling of both the plasma environment and dust dynamics will complement experiment through analysis using a comprehensive set of analysis techniques. The results will be used to predict the parameter space where particular wave modes can exist in experiment. Scientific/Technical Merit -- This research will be used to develop comprehensive theories on interparticle potential and grain-plasma interactions which determine the microscopic and macroscopic properties of complex plasma, including external forces acting on the particles, the fundamental interactions between the particles, thermodynamics, and self-organization of the system. The results will directly impact research in astrophysics and planetary science (dust in protoplanetary disks; charged dust on the surface of airless bodies, such as the moon and asteroids; planetary ring systems), atmospheric physics (noctilucent clouds), fusion research (dust contamination, operating stability), materials physics (soft matter, metastable states, complex fluids, granular matter), and advanced manufacturing (semiconductor, etching, and deposition systems). Broader Impacts -- Students will be engaged in this research allowing them to develop computation and modeling skills; obtain practical experience with vacuum systems, instrumentation and diagnostics; and improve technical writing and presentation skills. The research collaboration proposed under this project will also allow graduate and undergraduate students to travel between participating institutions, establishing their connection to the larger plasma community.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Impact (Primary) Numerical (Baylor) - A new code has been developed to model the interactions of dust with flowing ions. This allows the dust charging and dynamics to be self-consistently calculated, including the effects of ion wakefield on the dust-dust interactions and charging. While many features to be implemented within the code are specific to the PK4 discharge, the code is robust and can be applied to many different dusty plasma experiments by changing the boundary conditions governing the ion flow and external confining potentials. The code allows users to explore the parameter space of experimental conditions to determine those which are favorable for a particular dust configuration, such as one-dimensional strings. Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - The results of the PIC/MCC simulations of the PK-4 gas discharge predict the presence of fast ionization waves in both the stationary dc discharge and during the polarity switching. These rapid features remained undetected in the experiment due to the lack of fast diagnostics but are apparently of key importance for the formation of the observed dust particle chains. These findings provide the necessary foundation for the design of future experimental campaigns. Research Impact on Other Disciplines: The field of complex plasmas is now considered the plasma state of Soft Matter. As such, these results will push research in this field forward. Research Impact on Human Resources: Baylor - To date, three undergraduate students have been involved in the development of various codes at Baylor (one of these students is a member of an under-represented minority). Two graduate students (one female) are starting to use the code in their graduate research. The students gained experience in testing and developing code, learned how to program using CUDA, and sharpened their skills in C++ and Matlab. Each of these programming languages is listed among the top twenty programming languages for 2017 by the Institute of Electrical and Electronics Engineers (IEEE). In addition to research, the students are also gaining experience in technical writing and speaking. Research Impact on Physical Infrastructure Experimental (Baylor & Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - Motivated by the new insights provided by the numerical simulation of the gas discharge plasma, a new PK-4 replica system is being built at the Hypervelocity Impacts & Dusty Plasmas Lab (HIDPL) laboratory at Baylor University and the Wigner Research Centre for Physics in Budapest, Hungary. The primary aim is to provide experimental verification of the numerical results and the discovery of new details that need further explanation to help the design of future experimental campaigns on the PK-4 facility at the ISS. Research Impact on Computing Resources Baylor - The IonWake code is a new tool which will prove useful for many years to come. Computer codes which provide new capabilities are used for ten to twenty years (or more) and provide a research platform for many future

graduate and undergraduate students. As an example, over a period of ten years previous codes have been used by about 20 undergraduate students and 3-5 graduate students.

Research Impact Beyond Science & Technology: The impact of this project on graduate students, undergraduate students, and K-12 students will impact the overall science literacy of the voting public, making an impact beyond the bounds of science and engineering alone.

Project Goals
The goal of the proposed research project is to investigate the underlying physics behind self-ordering of interacting complex plasma dust particles into field-aligned chains in a microgravity environment.

This goal will be accomplished by pursuing the following specific objectives:

1. Establishing the complex plasma parameters required for the formation and stability of extended field-aligned dust particle chains,

2. Measuring the grain-grain potential along and perpendicular to the ion flow direction, and

3. Studying interchain interactions and their role in the structural properties of an ordered complex plasma system.

Project Progress (Year One)

Major Activities

Experimental (Baylor) - A ground 'mock-up' of the PK-4 has been constructed at Baylor University and is now undergoing initial calibration and christening tests. The instrument tube is identical to the one currently in use at the University of Iowa (John Goree) and as close to the tube employed by the European Space Agency/German Aerospace Center (ESA/DLR) in their ground model as possible. Both first light and first data have been achieved.

Experimental (Russian Academy of Sciences) - An application of the correlational approach to the problem of experimentally exploring the wake-mediated nonreciprocal interactions in complex plasmas has been developed. Simple analytic equations have been derived, which allows calculation of the derivatives of the nonreciprocal interaction forces in a strongly coupled many-particle system as well as the gradients of the external field. These calculations use data on time-averaged correlations of particles displacements and velocities, which are easily measured in an experiment.

Initial experiments with two dust particles in an RF discharge have been conducted to test the proposed method for determining the wake-mediated forces of interparticle interaction in complex plasmas.

Additionally, the development of amplitude instability in a two-particle Yukawa system for a wide range of values in screening parameter, friction coefficient, and gradients of external electric field has also been examined.

Experimental (DLR) - Additionally, in order to better prepare for the future experimental campaigns an extensive search of available experimental data has been conducted to look for conditions conducive to the formation of particle strings.

Particle strings have been clearly seen in PK-4 campaign data with the following three findings of relevance to the present project: (1) a regime has been found where particle strings form, (2) the coexistence of strings made of particles of two different sorts has been proven, and (3) the effect of the auxiliary RF discharge on particle string formation has been established. Observations like this will help us to identify experimental conditions for future experiments on the ISS.

Numerical (Baylor) - An entirely new code has been written to model the ion-dust interactions within the PK4 discharge. The code is written in C++ with CUDA to make use of the parallel processing power of Graphical Processing Units (GPUs).

The IonWake simulation is a MD simulation of the ion trajectories which includes ion-ion accelerations, ion-dust accelerations, ion insertions (assuming a shifted Maxwellian velocity distribution incident on the simulation boundary), and ion deletions due to leaving the simulation boundary or impacting the dust particle. The algorithm has been tested and used to re-create the ion number density and potential plots within a spherical simulation region.

The IonWake code has been updated to include dynamic charging of the dust particles and boundary conditions for a cylindrical cavity within the background ion flow, which is an optimal configuration for modeling strings of dust particles. The dust dynamics are self-consistently calculated within the code by performing 1000 ion timesteps to achieve an equilibrium ion configuration and dust charge. The forces acting on the dust grains from ions, dust, and external confining field are then calculated, and the dust is advanced on its own timestep, generally 104 ion timesteps. A new ion configuration and dust charge is then calculated over the next 103 timesteps.

Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - A particle in cell code including Monte Carlo collisions (PIC-MCC) to model gas discharge simulations is now on-line. It is capable of accurately describing the gas-phase processes and relevant surface processes on the level of elementary electron-atom and ion-atom collisions. The simulation includes the collisional cross section set published by Biagi (v7.1), which includes electron – neon elastic scattering, ionization and excitation to 9 different (including 2 metastable) energy levels. Ion – atom collisions include isotropic elastic and symmetric charge transfer collisions. Metastable density distribution calculation and Penning ionization are included in the model. Surface processes include electron reflection and ion induced secondary electron emission from the electrodes and surface charging on the glass wall. The results of the simulations provide quantitative information of the charged particle densities, fluxes, electric field and energy distributions as well as distribution of specific collisional events, like the ionization and excitation source functions.

Specific Objectives

Numerical (Baylor) - Development of an IonWake code allowing the ion density and wakefield structure to be determined for a chain of dust particles embedded in an ion flow. This would allow the grain-grain potentials to be calculated for a variety of chain configurations and ion flow speeds.

Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - The specific objective of this line of research is to enhance our understanding of the dynamical processes in the gas discharge environment, which

is primarily responsible for the dust particle chain formation, and the quantitative computation of relevant plasma parameters at the position of the dust cloud.

Experimental (Russian Academy of Sciences) - Development of a new method for determining the wake-mediated forces of interparticle interaction in complex plasmas and investigation of the amplitude instability in a chain-like system of dust particles suspended in a plasma with flowing ions.

Significant Results to Date

Numerical (Baylor and Wigner Institute) -The parameters characterizing the plasma environment at the position of the dust particles have been determined by means of numerical gas discharge simulations under experimentally relevant conditions.

Task Progress:

Experimental (Baylor) - A multi-chain structure, trapped in the PK-4 using a DC polarization switch has been examined. The spontaneous motion of the particles tracked has been determined by finding their positions and velocities in each continuous frame. Since the chains trapped have a shear movement relative to each other (center chains move in one direction and chains on the edge in the other), investigation has been limited to particles within a single chain and the overall movement of the chain has been eliminated by considering the relative velocity using a specified particle as the reference. From Fourier transformation the dispersion relation (phonon spectrum) for longitudinal and transverse DLWs have been obtained and shown for three chains at three different radial positions. The background noise (overall vibration) caused by the system not being isolated has been shown to only affect the spectrum at k = 0 -- therefore, it does not affect investigation of the overall shape of the dispersion relation.

Numerical (University Of California, San Diego-UCSD) -

I. Striations in simulations of PK-4 discharge

The UCSD Principal Investigator (PI) collaborated on interpreting plasma structures observed in the PK-4 discharge simulations conducted by the Wigner Research Centre for Physics (WRCP) group. The structures were interpreted as striations or ionization waves. The observed peaks in the electric field at a particular position as a function of time were also interpreted as moving ionization waves. It was also pointed out that the plasma parameters used in the simulations appeared to correspond to a regime where experimental data in the literature on noble gas discharges indicate that striations can occur.

II. 1D dust lattice waves in an electrorheological dusty plasma

It has been shown that 1D chains of dust can form in "electrorheological (ER) dusty plasmas" in microgravity experiments under the action of a uniaxial ac electric field that leads to an attractive dipole interaction between the charged dust in addition to the repulsive Yukawa interaction (e.g., Ivlev et al., 2008, Phys. Rev. Lett.100, 095003). In PK-4 experiments, strings of dust grains have been observed under electric field polarity switching conditions (e.g., Pustylnik et al., 2016, Rev. Sci. Instr. 87, 093505, Figs. 12 and 17). In this case, the interaction between dust grains may possibly be of the ER form of a combined Yukawa and effective dipole interaction. These strings are observed to occur in bunches in PK-4, so a 2D or 3D description, including finite boundary conditions, would probably be needed to describe waves in such a system. The UCSD PI envisions a road map to this goal, beginning with a more detailed description of waves in a 1D string, and then adding interactions between strings.

As an initial step, the UCSD PI is extending a previous analysis of lattice waves in a 1D ER dusty plasma (Rosenberg, 2015, J. Plasma Phys. 81, 905810407) to include the next nearest-neighbor interaction, assuming one can indeed model the binary interaction as an ER interaction as given in Ivlev et al. (ibid). Besides finding the dispersion relation of the normal modes, conditions under which the dispersion curve may peak inside the first Brillouin zone are being investigated.

As another initial step, the UCSD PI plans to extend the 1D lattice wave calculation to the strongly coupled liquid phase. This would require simulation data and/or experimental data on the pair correlation function. Preliminary estimates of possible dusty plasma parameters for the liquid phase have been provided to the WRCP group.

III. Electromagnetic wave scattering from 1D dust chains

Motivated by the observations of 1D string-like structures of dust grains in microgravity experiments, the UCSD PI is beginning to study the scattering of electromagnetic waves from 1D chains of dust particles, with a focus on predicting conditions for Bragg scattering. The Bragg scattering would be in the THz range (roughly 30 micron – 3 millimeter) because dusty plasmas have inter-grain spacings in this range. The UCSD PI has begun to use a publicly available scattering code to compute the scattering matrix elements, given the number of grains in the chain, the grain radius, the inter-grain spacing, and the optical properties of the grain material. The goal is to explore whether there are feasible conditions (e.g., dust size, spacing, materials) under which Bragg scattering from a 1D dust chain might lead to a type of photonic crystal with a stop band in the THz. This is motivated by reports of tunable photonic crystals in the visible from systems of chain-like structures of charged, super- paramagnetic colloids in a magnetic field (e.g., Ge et al., 2007, Angew. Chem Int. Ed. 46, 7428).

Experimental (Russian Academy of Sciences) - A general approach to the correlational analysis of Brownian motion of strongly coupled particles in open dissipative systems is under develpopment [Lisin et al., Physics of Plasmas 25, 013702 (2018)]. This approach can be applied to the theoretical description of various non-ideal statistically equilibrium systems (including non-Hamiltonian systems), as well as for the analysis of experimental data. Using the correlational approach, we developed a new method for determining the wake-mediated forces of interparticle interaction in complex plasmas. In the examples of numerical simulations, we demonstrate that the proposed method could be an effective instrument in exploring the wake of a dust particle in a plasma. Unlike the previous attempts to study the wake-mediated interactions in complex plasmas, our method does not require a special design of the experimental setup or any external influences on a system, pre-measurements and any assumptions about the form of interaction. It is based on Brownian motion analysis only and can be used to study many-particle chain-like structures in complex plasmas.

We also showed the importance of taking dissipative and random processes into account, without which consideration of a system with a nonreciprocal interparticle interaction as linearly coupled oscillators leads to significant errors in determining the characteristic frequencies in a system. We obtained that dissipative and random processes determine the minimum value of the force derivative to which the particle “reacts.”

	The proposed method was tested in the experiments with two dust particles levitating in the sheath region of an RF discharge. We also investigated the amplitude stability in a two-particle Yukawa system, i.e., the stability to any-size deviations (not just small ones) of particles from their equilibrium positions [Lisina et al., Physics of Plasmas 24, 113705 (2017)]. We presented an analytical approach for the amplitude instability prediction based on the determination of an inflection point of the system potential energy. Our new approach can also be an effective tool for studying amplitude instability in systems of interacting particles suspended in plasma with directed ion flow which adds to the interaction forces a wake-mediated attractive force. Key Outcomes Numerical (Institute for Solid State Physics and Optics, Wigner Research Centre for Physics) - Our first results predict the presence of strong ionization waves in the stationary discharge and the appearance of an ionization tsunami during the polarity switching as it is applied in the PK-4 experiment. The computed plasma parameters serve as input for the dust grain charging models and the calculations of the inter-particle interactions.
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